Linear Regression

QLS–MiCM Workshop

Learning Goals

After this module, you will be able to:

- Fit linear regression models in Python.
- Interpret coefficients.
- Compare simple vs. multivariable models.
- Evaluate model fit.

Why Linear Regression?

A common model in clinical research because:

- Interpretable coefficients.
- Well suited for tabular data.
- Often enough!

What Linear Regression Does

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$$

- Predicts a **continuous** outcome (Gaussian).
- coefficients (β_p): change in y for unit change in x_p.
- Assumes relationships are additive.

Step 1: Start Simple

We begin with a model that uses **one predictor**.

$$y = \beta_0 + \beta_1 x_1$$

This helps us:

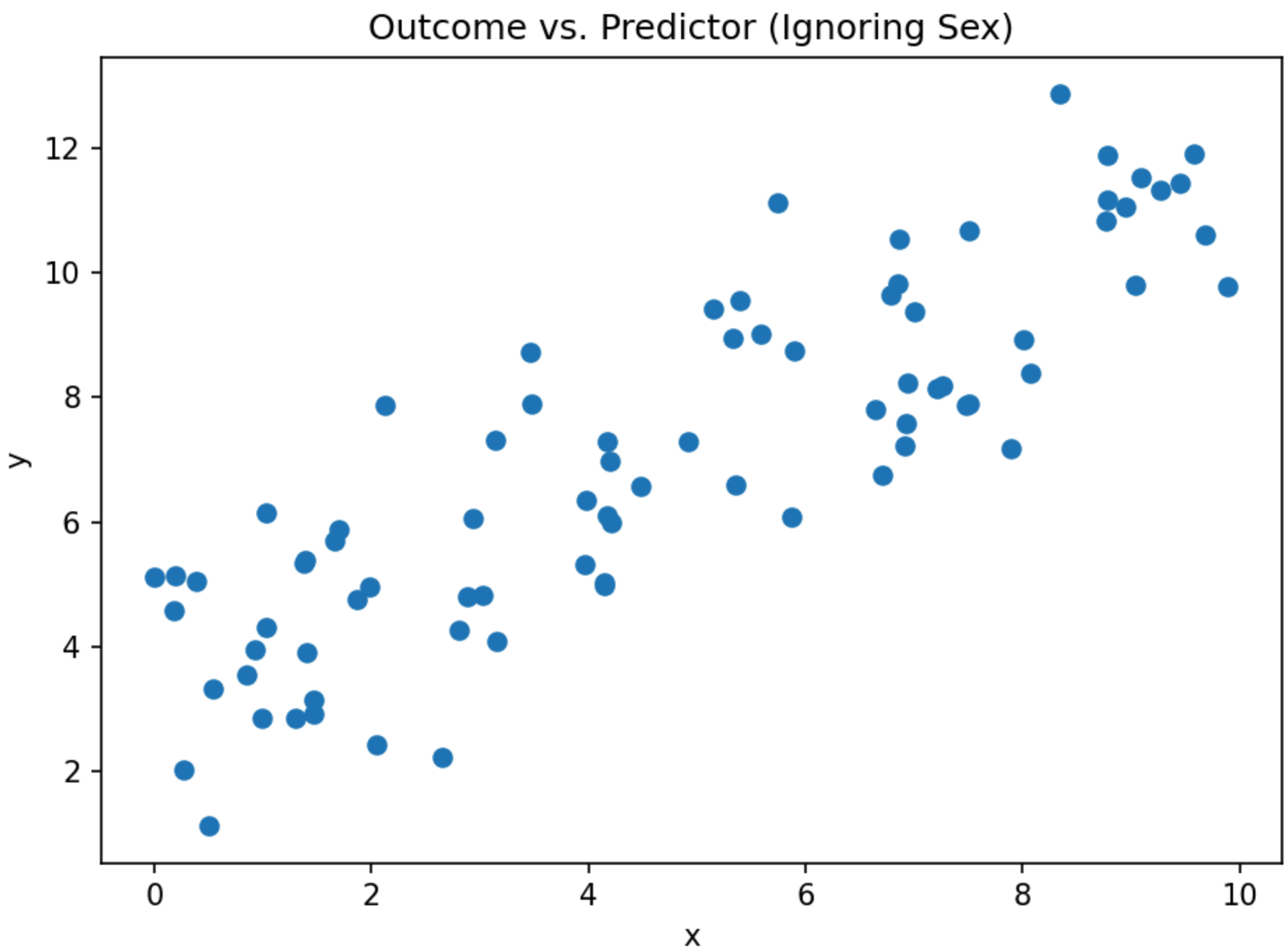
- Build intuition.
- Understand effect sizes.

Step 2: Add More Predictors

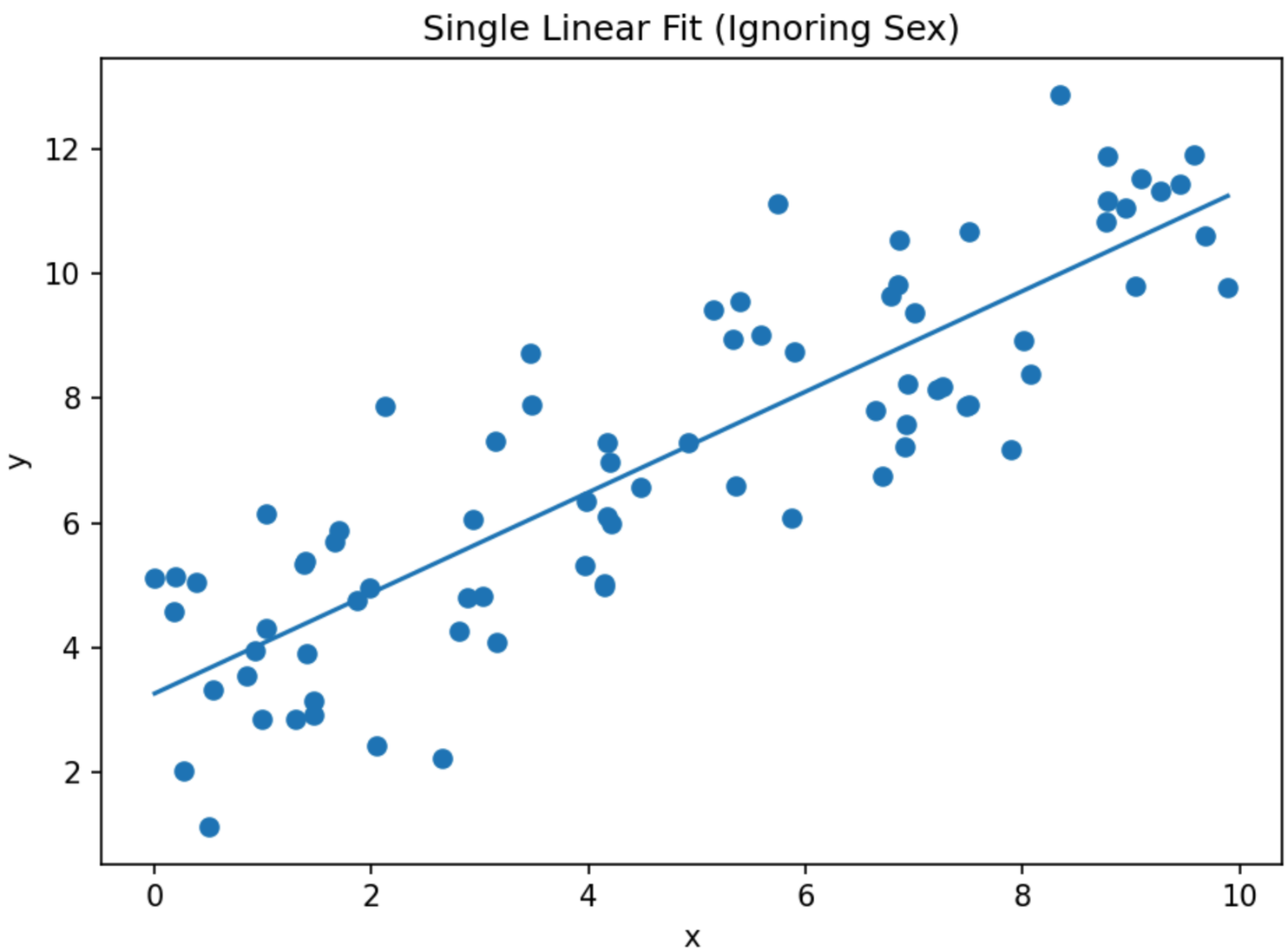
Questions to consider:

- Do coefficients change?
- Does performance improve?
    - Is that improvement useful?

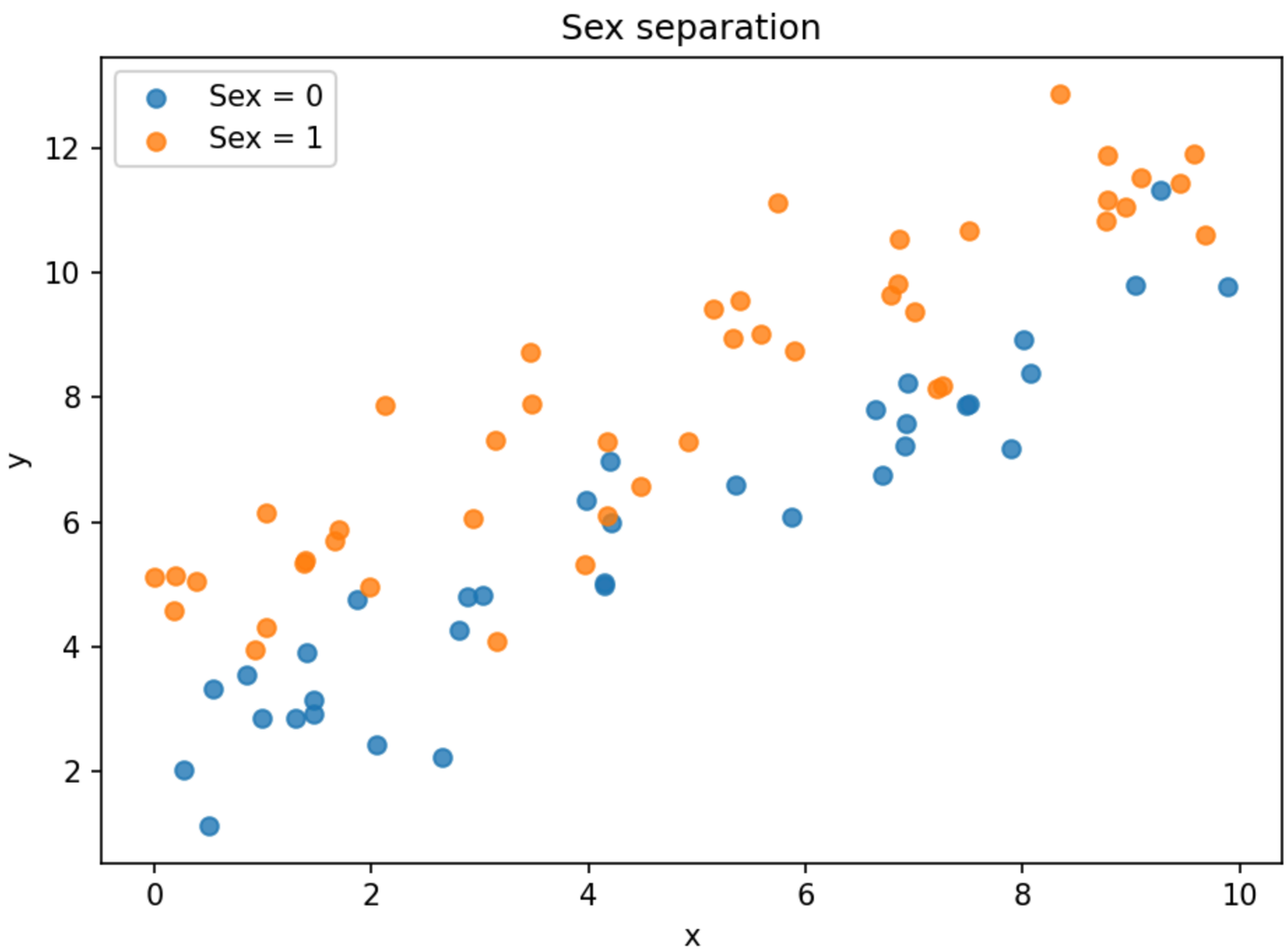
Example 1: Outcome vs Predictor (Ignoring Sex)



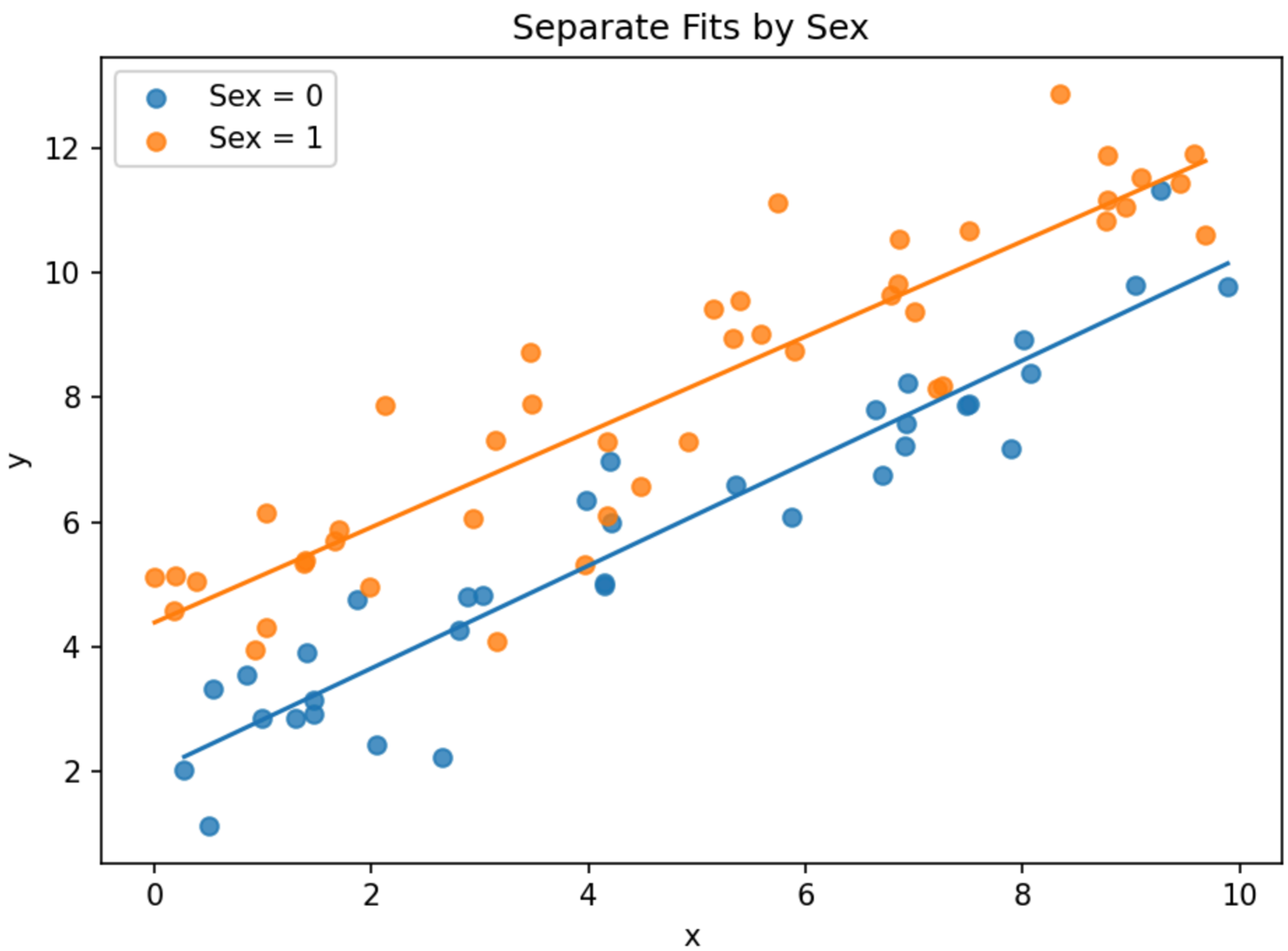
Example 2: Single Linear Fit (Ignoring Sex)



Example 3: Sex as a Categorical Variable



Example 4: Separate Regression Lines by Sex



Step 3: When Predictors Relate to Each Other

- How does the model handle related predictors?
- Are coefficients stable?
- Does interpretation become easier or harder?

Evaluating the Model

You will compute:

MAE: Average prediction error.

R^2: How much variation in the outcome the model explains.

- What is “good enough”?
- Do more predictors always help?

Understanding Uncertainty

With `statsmodels`, you will look at:

- p-values.
- significance.

Consider:

- What makes a coefficient reliable?
- How confident are we in each effect?

Interaction Terms

You will also try adding an interaction term.

This shows how linear models can:

- represent relationships between predictors.
- become more expressive.

And raises questions like:

- When is this helpful?
- When might it complicate interpretation?

What You Will Do

In the notebook, you will:

1. Fit a model using **one** predictor.
2. Fit a **multivariable** model.
3. Add an additional predictor.
4. Explore how the model behaves.
5. Compare evaluation metrics.
6. Examine uncertainty in coefficients.
7. Reflect on your findings.